

Writing Research Papers

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For researchers turning technical evidence into a paper that readers can follow and check. Use this guide to identify the strongest claim supported by the evidence, then shape the paper around that claim without hiding uncertainty or flattening the authors' voice.

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Abstract

A technically correct paper can remain difficult to evaluate when its central claim is separated from the evidence that supports it and the conditions under which it holds. Effective paper writing begins with a claim–evidence map, then carries those relationships into the order of sections, figures, and sentences. This guide shows researchers how to make that argument visible without changing its technical meaning or flattening the authors' voice. The finished paper should let a careful reader identify the central claim, inspect its support, and understand the conditions under which it applies.

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1 Start with the paper's question

A paper becomes difficult to write when its authors have not fixed its central question and claim. Polishing cannot settle either one. Before drafting, state the scientific problem and the paper's answer in ordinary language.

1.1 Write the paper's contract

Write four sentences before writing any section:

1. **Problem.** What practical or scientific problem does the paper address?
2. **Gap.** Why does the strongest current method or explanation leave that problem unresolved?
3. **Claim.** What does this paper establish?
4. **Boundary.** Under which conditions does the claim hold, and which important conclusion or question remains outside its support?

These sentences form a contract with the reader. The introduction may motivate the problem and the body may develop the details, but neither should quietly change the contract. The boundary sentence matters as much as the central claim. It prevents a result in one regime from becoming a claim about every regime. It also keeps a measured effect separate from the explanation proposed for it.

A useful one-sentence test is:

We study *[problem]* because *[current limitation]*; we show *[central result]* under *[scope]*, while *[limitation]* remains open.

The finished sentence need not appear in the paper. Its purpose is to expose disagreement among the authors. If two plausible versions imply different experiments or different section orders, settle that disagreement before line-editing the introduction.

Match the intended audience to the reach of the evidence. State the new understanding the paper offers that audience, then look to adjacent fields for a useful framing. That comparison can strengthen the argument without inflating the claim.

1.2 Map claims to evidence

List every claim required for the paper's conclusion to hold. Then assign each claim a visible piece of evidence. A theorem supports a formal statement under its assumptions. A controlled comparison supports a comparative claim. An ablation isolates a design choice. A mechanism study tests an explanation. A qualitative result can make a failure mode concrete, but it rarely establishes frequency on its own.

Claim	Evidence needed	Boundary to state
Headline result	Theorem or controlled benchmark	Assumptions, tasks, metrics, and uncertainty
Mechanism	Diagnostic intervention or measured intermediate behavior	Alternative explanations that remain possible
Practical value	Cost, deployment result, or stress test	Resources and settings not yet tested

This map should reveal weak links. A contribution with no evidence is a proposal, not a result. Evidence that does not support the central argument may still belong in the appendix, but it should not dictate the main paper. When a claim depends on a missing control, record the gap instead of drafting around it.

1.3 Choose the level of the contribution

Papers make different kinds of contributions. A method paper may show that a new procedure resolves a concrete tradeoff. A mechanism paper may overturn a popular explanation through controlled tests. A theory paper may identify a failure and prove when it disappears. Some papers combine these forms, but one usually carries the main argument.

Defining a constrained regime can also be a contribution. A careful problem definition may expose assumptions and design choices that abundant data or a looser formulation hides. In that case, say that the paper defines the regime, tests a recipe, or identifies necessary conditions. Do not say that it solves the broader problem.

Name that primary contribution early. It determines the evidence that deserves space and the content of Figure 1. It should also govern the title and abstract. A paper centered on explanation should not market itself as an algorithm paper because it happens to introduce a diagnostic model. A paper centered on a theorem should not bury the regime in which the theorem applies.

2 Settle the argument before drafting

The section list should follow the logic of the evidence. A fixed outline can hide the paper’s strongest result under routine setup or give equal space to claims of very different importance. Decide the reader’s prerequisites and the evidence for each step. Section titles and paragraph topics follow from that sequence.

2.1 Open with the scientific bottleneck

Begin with the scientific goal and the resource or assumption that blocks it. Do not let the nearest acronym or prior method define the motivation unless a reader must understand that method to see the problem. A method-specific opening can make a broad question look like a narrow competition between techniques.

Name the scarce and available resources. For example, a paper about robot adaptation might study a setting with few expert demonstrations but ample autonomous interaction. That framing identifies the demonstration bottleneck before choosing imitation learning, in-context learning, offline learning, or online reinforcement learning as the response.

2.2 Choose the structure from the evidence

The example papers use three recurring architectures.

Empirical method papers. *DPPO* [Ren et al., 2024] and *OGPO* [Patil et al., 2026] begin with an operational bottleneck, identify a tradeoff in current methods, and introduce a method that changes that tradeoff. Headline comparisons establish the main result. Later experiments test the proposed explanation. They also identify performance limits and test deployment. This architecture works when performance is the central claim but mechanism and reliability determine the result’s credibility.

Empirical mechanism papers. *Much Ado About Noising* [Pan et al., 2025] organizes the paper around competing explanations. Controlled tests first rule out explanations that do not account for the observed performance. A minimal replacement then isolates the components that do matter, and later experiments examine their mechanism. This architecture fits a paper whose contribution is a corrected understanding. The negative and positive findings must remain distinct: rejecting one explanation does not prove another.

Theory-led papers. *Action Chunking* [Zhang et al., 2025] starts from a failure in continuous-control behavior cloning and studies two interventions in different regimes. The formal results establish each intervention’s sufficient regime, while experiments test the predictions that matter outside the model. This architecture fits a paper that moves from a failure or impossibility to guarantees under stated conditions. The regime should appear before the guarantee, both in prose and in section titles.

Treat these as shapes. A new paper may need a different order. All four examples open with a concrete problem. Each then exposes a mismatch in current thinking and resolves it as far as the evidence allows. Each section should resolve a question created by the previous one.

2.3 Present the decisive result first

In an empirical paper, show the headline result before low-level ablations. The reader needs to know that the method changes the outcome before learning the relative importance of regularizers and schedules. Mechanism studies should follow the performance claim unless mechanism is itself the main claim.

In a theory paper, present the motivating failure before the intervention. State assumptions before the result that uses them, but avoid a long block of notation that will not be needed for several pages. Definitions are easier to absorb when they arrive close to the question they answer.

Related work can appear early or late. Place it early when the reader must distinguish the method from close alternatives before understanding the formulation. Place it after the main results when a literature survey would delay the paper’s question and answer. In either case, organize related work by technical distinction. A chronological list of papers does not identify the unresolved technical point.

2.4 Draft the evidence before the introduction

The order of the paper need not be the order of writing. A reliable drafting sequence is:

1. settle the claim–evidence map;
2. draft the central figures, tables, theorem statements, and captions;
3. write the results that interpret that evidence;
4. write the setup and method needed to understand those results;
5. write the introduction once the contribution hierarchy is stable;
6. write the abstract and title last.

This sequence reduces a common source of churn. An early abstract often commits the paper to claims unsupported by the final evidence. An early introduction can also turn provisional organization into a constraint. The abstract should explain the argument supported by the finished evidence, not tour the paper’s sections or describe the paper the authors first imagined.

3 Build sections around the reader's questions

Write with a particular reader in mind. Each section should answer a question the paper has raised, and each paragraph should carry one step of that answer. This gives coauthors a practical test for duplication. Combine sections that answer the same question. If useful details interrupt the argument, move them to the appendix. A target page count cannot make either decision.

3.1 Use the introduction to state the argument

Open on the concrete task and the failure of current methods or theory. Narrow quickly to the exact failure. The reader should understand the tension before seeing the solution.

Then state the central insight in ordinary language. The method name and technical machinery can follow. A reader who does not yet know the notation should still be able to state the change and its importance.

The problem statement should open the paper's design questions. A useful form names the starting object, the scarce resource, the resource that remains available, and the outcome to explain. The rest of the introduction can then separate questions about the representation, data, algorithm, and evaluation instead of presenting one method as the entire answer.

Present contributions as supported claims:

Sample efficiency. We compare the method with [baselines] under [controlled setting] and find [result] on [tasks or regimes].

The heading names the claim, while the sentence gives the comparison and evidence. “We introduce a new algorithm,” “we run extensive experiments,” and “we provide theory” name author activities. They report effort, not findings.

For a paper that defines a constrained regime, the contribution sequence may include the problem definition, a tested recipe, necessary conditions, a characterization of the success and failure regimes of existing approaches, and a new evaluation criterion. Each item still needs a result. List the concrete findings before explaining their broader context.

Close the introduction with the paper's conceptual takeaway and its scope. Do not repeat the same contribution in several formats. Choose one primary signposting device and let the other elements add information.

3.2 Place setup where readers need it

Place each definition or assumption close to the first substantive result that needs it. Keep standard preliminaries short. Spend space on a distinction that later carries a claim, especially when two methods use similar notation but optimize different objects.

A method section should explain the conceptual move before presenting every implementation choice. Start from the limitation identified in the introduction. Identify the changed computation and its purpose. Then give the formal definition and practical variants.

A theory section should state the regime and result before a long derivation. Readers need the theorem's purpose and target question. Keep the main proof idea near the theorem when it carries scientific insight. Move routine algebra and complete technical cases to the appendix. Never promote an intuition to theorem language, and never omit a condition because it makes the statement less elegant.

3.3 State the question behind each experiment

Begin the experimental section with the questions the experiments will answer. Explain the distinction tested by each task and metric. Then justify the baselines and state the controls. A comparison cannot support a clean mechanism claim when the architecture, training objective, and initialization all change at once.

The usual order is:

1. evaluation regimes and controlled comparison rules;
2. headline performance;
3. stress tests or deployment;
4. mechanism studies;
5. ablations of design decisions;
6. efficiency, failures, and limitations.

Change this order when the scientific question requires it. A mechanism paper may need a controlled falsification before any broad benchmark. A paper about sample cost should report environment interactions before wall-clock convenience. The metric must match the claim.

State a result as a comparison. Name the metric and setting before offering an explanation. Separate a performance comparison from a mechanism test. “Method A outperforms Method B” and “component X causes the improvement” are different claims and need different evidence.

Study each enabling condition separately. Vary the representation, task coverage, update algorithm, and initial policy performance. Initial success may not predict adaptability. Treat that difference as an experimental question, not as an interpretation added after the results.

3.4 Use related work to draw technical distinctions

Group prior work by the technical choices that matter for the current paper. For each group, explain its aim and the assumption that separates it from the current paper. End with the point that remains unresolved. Cite the closest alternatives directly. Broad claims such as “few works consider this setting” require a careful search and a clear definition of the setting.

Treat a related method as evidence about a scientific distinction, not merely as an opponent. Explain its complementarity with the current approach and name the different resource or assumption behind each method. For readers outside the subfield, describe the specialized method before relying on its acronym. A citation supports that explanation; it does not replace it.

Related work should not become a second introduction. It should help the reader place the paper’s contribution accurately, including work that weakens the novelty claim. A narrow, correct distinction is more credible than a broad claim of being first. Cite distinctive language or framing that comes from another paper. Do not disguise borrowed phrasing by rotating synonyms.

3.5 Separate implications from limits in the discussion

Use the discussion to interpret the evidence at the right level. State the regimes in which the result should change practice or understanding. Then name the limitation that most affects that conclusion. Untested environments may narrow the result’s breadth; a strong assumption may change where it applies.

The conclusion should not introduce a new contribution. It can state the central result, its consequence, and the open question that now matters. A generic recap weakens the ending. End on the most precise supported implication or boundary.

3.6 Make the appendix usable

A long appendix benefits from a table of contents and an order designed for readers. Put complete proofs and the details needed for reproduction there. Practitioner guidance and qualitative failure analysis also belong in the appendix when they help readers reuse the work.

The main paper must still support its central claim. Do not move a decisive control or a condition that changes the theorem's meaning to the appendix merely to save space. Move details that permit verification and reuse while keeping the argument itself visible.

4 Make the figure sequence support the argument

Readers often meet the evidence through figures before they read the full paper. The figure sequence should make the same argument in shorter form. A reader who scans the title, abstract, and figures should understand the problem and the main result. If the paper proposes an explanation, the figures should show the evidence for it.

4.1 Use Figure 1 to summarize the paper

Figure 1 should identify the new intervention and show its main effect. If the paper makes a mechanism claim, the figure should also show the evidence that motivates it. Most first figures pair a method diagram with one headline result. Add a task image or mechanism sketch only when it supplies necessary context.

The example papers make different choices because their claims differ. *Action Chunking* places its two interventions in one conceptual frame. *DPPO* joins the nested decision process to robot tasks and the properties it studies. *Much Ado About Noising* uses a component taxonomy to reject prevailing explanations and introduce a minimal replacement. *OGPO* places the algorithm beside sample-efficiency curves, then uses its final panel to motivate the mechanism study. In each case, the first figure reflects the paper's main claim rather than a standard figure template.

Design Figure 1 after the claim–evidence map is stable. If the figure requires a claim that is not tested in the paper, change the figure or collect the missing evidence. If the figure cannot state the contribution without several paragraphs of caption, the contribution hierarchy may still be unsettled.

4.2 Use later figures to answer distinct questions

A later figure should resolve one question left open by Figure 1. Useful figure types include:

Concept figure.

Makes a theoretical object, information flow, or mechanism visible.

Controlled comparison.

Changes one disputed component while holding the relevant alternatives fixed.

Headline benchmark.

Establishes the main comparison across representative regimes.

Mechanism figure.

Shows trajectories, distributions, gradients, or intermediate computations that bear on an explanation.

Qualitative sequence.

Displays an ordered success or failure with the event of interest marked.

Ablation.

Removes or varies one design decision and reports the resulting change.

Do not ask one figure to answer every question. A benchmark grid becomes hard to read when it also contains the method diagram, ablation, and mechanism claim. Separate figures let the prose distinguish performance from explanation.

4.3 Keep colors and labels consistent

Assign colors by meaning and keep those assignments stable. Use neutral gray for context and most baselines, then reserve one or two accent colors for the method and the main alternative. Learning curves should state the uncertainty shown and the number of seeds or trials. Direct labels are often easier to read than a distant legend. Panel headings should name the comparison, not merely “left” and “right.”

Diagrams need a visible direction of travel. Give each arrow one meaning and state that meaning in the caption. Task images can ground an abstract diagram, but they should show the actual task or failure under discussion. A decorative image does not supply evidence.

Inspect every figure at its final column width. Check every label and annotation at that size. Lines must remain distinguishable on a printed page, including in grayscale and under common forms of color-vision deficiency.

4.4 State the evidence in each caption

Start the caption with the figure’s claim or purpose. Walk through the panels in reading order. Then report the comparison and main observation before giving the narrowest interpretation the figure supports. Point to setup details when they are not visible nearby.

A useful caption order is:

1. claim or purpose;
2. panel-by-panel setup;
3. comparison and metric;
4. main observation;
5. narrow interpretation or caveat;
6. pointer to complete details.

The caption and results paragraph should agree. A caption cannot say that an ablation “demonstrates the mechanism” while the text calls it suggestive. Define nonstandard symbols and uncertainty either in the caption or in the adjacent prose. The reader should not need to search the appendix for the meaning of an error band.

5 Write direct technical prose

Good technical prose lets the argument remain visible at sentence scale. Put the main subject and verb near the front. The example papers use first-person verbs to mark author actions. “We compare” names an experimental choice, while “we find” reports its result. They reserve “we hypothesize” for an interpretation. Use the technical object as the subject when the behavior belongs to the object.

Author action: We compare the two methods under a shared initialization.

Technical behavior: Action chunking limits compounding error under the stated stability condition.

Direct prose does not require short sentences. A long sentence works when every clause serves one relation. A short sentence fails when it adds drama but no claim.

Read a revised sentence as a whole. A local repair can preserve the grammar while making the relation stiff or indirect. Recast the sentence when a relative clause or passive construction sounds mechanical. No grammatical shape is worth keeping for its own sake.

5.1 Put conditions next to the claims they constrain

A fronted clause should change the interpretation of the main claim:

Although the estimator is biased, its error does not grow with the horizon in this regime.

When the system is open-loop stable, action chunking gives the stated bound.

Because the critic supplies the terminal reward, the method can reuse environment data off policy.

Openings such as “while” and “although” become clutter when they provide generic scene-setting. If the clause does not change the main statement’s interpretation, lead with the statement.

5.2 Make one main inference per sentence

A sentence may contain a result and its essential condition or qualification. Split it as soon as a second inference enters. The method, result, and proposed explanation rarely belong in the same sentence.

OGPO reuses environment data through an off-policy critic. It spends additional computation on denoising trajectories instead of collecting new environment interactions.

The two sentences describe related parts of the same idea, but each states one main inference. This separation also makes technical review easier. A coauthor can challenge the data-reuse claim without untangling three other claims from the same sentence.

Every sentence should add information. Remove empty pointers such as “in this,” “this shows,” and “we can see that” when the reader must search for their referent. Keep a pointer when it names the object directly: “In this regime, the error does not grow with the horizon.” Include a familiar background fact only when it helps the reader understand a later distinction.

5.3 Develop one main inference per paragraph

Most technical paragraphs make three or four moves:

1. state the claim;
2. give the reason or technical distinction;
3. present the theorem, comparison, measurement, or example;
4. state the supported consequence or boundary.

Use only the moves required by the paragraph. A two-sentence paragraph can be complete. Vary sentence length and syntax so that paragraphs do not read as filled-in forms. End on a consequence, limit, or transition that creates the next question. A final sentence that merely repeats the first can usually be deleted.

Paragraphs should also have explicit agents. Write “we compare” instead of “a comparison is conducted,” and “the policy fails” instead of “a failure is exhibited by the policy.” Nominalizations hide the action and lengthen the sentence. Direct verbs leave more room for the scientific detail.

5.4 Use sentence starters while drafting

A sentence frame can make the paragraph’s purpose clear to the reader. Fill the frame with the paper’s actual objects and discard it when a more natural sentence appears:

We study [problem] in the regime where [constraint].

Existing approaches assume [resource]; we ask what is possible when [constraint].

To test whether [mechanism], we [intervention or measurement].

Compared with [baseline], [method] changes [metric] under [setting].

Within [scope], the result supports [inference] but leaves [boundary] open.

Do not start every paragraph from the same frame. The point is to make the claim, comparison, or question visible, not to give the paper a mechanical rhythm.

5.5 Report numbers before interpreting them

For a quantitative result, lead with the comparator and metric. Then report the value under the exact setting, including uncertainty, before interpreting it:

Compared with [baseline], [method] improves [metric] from [A] to [B] on [setting], averaged over [seeds or trials]. This supports [narrow inference].

Use the available numbers. “Substantially better” is less useful than the measured difference. Reserve “significant” for a defined statistical claim. If the paper reports uncertainty, name the quantity it represents. If two methods use different budgets, data, or initializations, expose that difference before the reader reaches the result.

5.6 Separate formal results from interpretation

State the regime before a theorem’s consequence:

Under [assumptions], [intervention] guarantees [bound].

Then interpret the statement in a separate sentence:

Theorem 2 bounds [quantity] by [rate]. Thus, in [regime], the error does not grow with [variable].

Preserve every quantifier and condition. Keep the asymptotic rate and scope exact. A clean sentence with a missing assumption is wrong. If an intuitive explanation is not part of the theorem, label it as intuition rather than folding it into the formal claim.

5.7 Calibrate mechanism language

Observation and explanation are different sentence types:

We observe [measured behavior]. We hypothesize that [mechanism] produces this behavior because [support].

Use “shows” or “demonstrates” for direct evidence. Use “supports” for partial evidence. Use “suggests” or “is consistent with” for an interpretation that remains uncertain. “We believe” does not supply evidence for a hypothesis.

A controlled intervention can support a causal statement when the relevant alternatives are held fixed. A visualization can reveal structure without proving cause; the same is true of a correlation. The verb should make that distinction visible.

5.8 Use contrast for real scientific distinctions

Contrast is useful when it identifies a real scientific distinction. Sample efficiency differs from cheap computation; performance evidence differs from mechanism evidence. State both sides in one clear relation.

Repeated formulas such as “This is not X. It is Y” or “not only X but also Y” make the prose sound rehearsed. They also encourage false binaries. Write the scientific distinction directly:

Current methods trade off sample efficiency and full-policy improvement.

When evidence rules out one explanation but does not prove another, say both facts without turning them into a slogan.

5.9 Replace abstract wrappers with verbs

Phrases such as “the mechanism behind X,” “the role of Y in Z,” and “the interplay between A and B” name a relationship without stating the result. Use a direct verb when the evidence supports one:

Abstract: We study the mechanisms behind *OGPO*’s stability.

Direct: We test whether conservative advantages prevent critic exploitation.

Sometimes the relationship itself is the question. State that question precisely. “We test whether stochastic supervision causes manifold adherence” is clearer than “we study the role of stochasticity in manifold adherence,” and it does not claim that the test has already succeeded.

5.10 Use lists only when the set matters

A sentence is not difficult because it is long. It becomes difficult when commas hold together several claims that have no hierarchy. The example papers often use long sentences to develop one contrast or causal relation. Follow that pattern.

Keep a full list when its members form a taxonomy, a set of assumptions, or a checklist. In ordinary prose, state the relationship among the items. Group them by their relation to the claim or split the sentence:

Packed: We evaluate stability, robustness, efficiency, and generalization.

Grouped: We first measure training stability and sample efficiency. Separate deployment tests measure robustness and generalization.

5.11 Keep technical terms stable

Introduce a technical term once, then give its operational meaning. Repeat the canonical term instead of cycling through synonyms. Use a pronoun only when its referent is obvious. Semantic LaTeX macros help keep method names and symbols stable across authors and sections.

Stable terminology does not require awkward word echoes. A sentence such as “Prior work builds on a prior policy” makes the reader assign two meanings to the same word in quick succession. Keep the technical term and rewrite the ordinary phrase. The same test catches repeated “in this” constructions and adjacent forms of the same verb.

Prefer one exact detail to a broad statement of importance. Shreyas puts the point sharply: “Details should feel like poetry: concrete, curated, and exact.” Treat that as a rule for selecting details, not for decorating them. Turn readers’ vague knowledge into an exact statement.

Good technical prose should feel precise and compressed. Its order can feel inevitable because each sentence creates the next question, not because the paper suppresses uncertainty or alternative explanations.

5.12 Use punctuation to expose the grammar

Use punctuation to expose the grammar. Commas mark real clause boundaries. Semicolons join closely related claims; colons introduce definitions or lists. Parentheses should contain secondary detail. Research questions belong in the prose when the question organizes the paper, not as a routine rhetorical setup. No sentence should depend on punctuation for drama.

6 Match every claim to the evidence

Technical editing begins with preservation. Keep every theorem condition and reported value intact. Preserve the notation, citations, and stated uncertainty. Do not make a sentence cleaner by deleting the fact that limits it.

The examples below are schematic. Their theorem conditions, task names, and numbers are invented; each pair isolates one editing choice.

6.1 Check the source before strengthening prose

Read the manuscript passage and the evidence it cites before editing a claim. Check a figure together with its caption, then consult the appendix for omitted conditions. A result that appears broad in the

introduction may depend on a narrow metric or a specific initialization. A theorem may use a stronger assumption than the surrounding discussion.

Never invent evidence or imply that planned work is complete. Preserve every theorem condition and quantitative result. Mark a missing citation, fact, or control for the authors. Words cannot replace the missing evidence.

Overstated: Theorem 3 shows that action chunking prevents compounding error.

Theorem 3 establishes: Under open-loop stability and bounded disturbances, Theorem 3 bounds rollout error for horizons up to H .

6.2 Separate hypotheses, planned tests, and findings

Write “we test whether” when the paper poses a question, “we hypothesize” when it proposes an explanation, and “we find” only after the evidence supports a result. “We believe” and “a natural follow-up” do not tell the reader the statement’s status as a hypothesis, planned test, or finding.

Make the source of a claimed capability explicit. Suppose a visual backbone is frozen during adaptation. Robustness to visual perturbations must then come primarily from the pretrained representation, but that reasoning is still a hypothesis until the evaluation tests it. A scoped statement is:

Because the visual backbone is frozen during adaptation, robustness to visual perturbations must primarily come from the pretrained representation. We test whether that robustness persists over the perturbation range considered here.

Name the environment, perturbation, task mode, or resource constraint that limits each result. Do not promote a planned comparison, an expected qualitative outcome, or an intuition from another field into a validated claim.

6.3 Check every promised contribution against the evidence

Each abstract claim should appear in the main paper. Contributions in the introduction should point to visible evidence. A figure caption should make the same claim as the paragraph that interprets it, and the conclusion should follow from results already presented.

Run these checks in both directions. A promised contribution with no evidence is an overclaim. A decisive result that never appears in the abstract or introduction may mean the paper’s contribution hierarchy is wrong. Repair the hierarchy before polishing sentences.

Unsupported: Method A consistently outperforms prior methods across diverse environments.

Supported by Figure 2: At 100,000 interactions, Method A improves mean return over Method B on two of three simulated locomotion tasks and matches it on the third (Figure 2).

6.4 State comparison rules

State a comparison well enough that a reader can reconstruct its logic. Name the comparator, metric, and task. Then record any difference in budget, initialization, or data source. Mechanism claims require tighter controls than broad performance comparisons because they aim to isolate a cause.

Report failures and negative results with the same precision as successes. They can identify the regime boundary, reveal sensitivity, or reject an explanation. Do not hide them in vague limitation language.

Missing comparison details: Method A improves success by eight percentage points.

With comparison details: Compared with Method B on Task Q, Method A raises mean success from 68% to 76% over 20 evaluation seeds. Both methods use 100,000 environment interactions and the same initialization distribution.

6.5 State limits where the result first appears

A limitation is useful when it changes the result’s application or interpretation. Name the condition that creates the limit and state the remaining support. If several limits matter, explain each one separately.

Do not use a broad limitations paragraph to excuse a broad claim. Narrow the claim where it first appears. Readers should not need to reach the final page to learn that the central result holds only in one setting.

Qualified too late: The method is robust to distribution shift. The limitations section later reveals that the tests changed only image texture in simulation.

Qualified up front: The method maintains its success rate under the simulated texture shifts tested here. Robustness to dynamics shifts and real-world observations remains unknown.

7 Keep the LaTeX source easy to inspect

The source tree can make revision easy or cumbersome. A clear project prevents duplicated text and accidental notation changes. It also lets several authors work without breaking a submission mode.

7.1 Use the entry point as an outline

Keep the top-level entry point short enough to read as the manuscript’s table of contents. It should show the build mode and manuscript order at a glance. Put prose in section files:

```
paper/
|-- paper-main.tex
|-- references.bib
|-- body/
|   |-- abstract.tex
|   |-- introduction.tex
|   |-- method.tex
|   `-- experiments.tex
|-- appendix/
|-- figures/
`-- preamble/
```

This guide preserves the complete preamble and title structure from OGPO [Patil et al., 2026] so that the original formatting remains inspectable. A new paper should keep the modular structure but remove unused project-specific commands. Source-specific macro sprawl makes later revisions harder to audit. Delete duplicate package loads and commented-out manuscripts.

7.2 Separate presentation from meaning

Stable package and layout choices belong in a style file. Put paper-specific notation in a commands file, including method names and operators. Keep author comments and colored revisions in a drafting file that can be disabled before release.

Keep venue `.sty` and `.cls` files unmodified. Load the venue style once, before project-specific styling, and record its version. Do not hide notation, spacing fixes, or author comments inside vendor code.

Use macros for semantic objects:

```
\newcommand{\method}{\textsc{Method}\xspace}
\newcommand{\state}{s}
\DeclareMathOperator*{\argmaxop}{arg\,max}
```

A method rename should require one edit. Avoid macros that conceal a whole sentence or arbitrary formatting. Source should remain understandable without opening several files for each line.

7.3 Use each toggle for one build distinction

Build toggles are useful when a release changes presentation. Use them for anonymous and public author blocks or arXiv metadata. They can also control acknowledgments, appendix length, and venue-approved theorem styles. They should not select different claims or experimental values. Two build modes with different science will drift.

Keep toggles near the top of the entry point. Build every supported mode before release. The public and anonymous versions may look different, but they should make the same scientific claims.

7.4 Keep figure sources reproducible

Store final vector diagrams and plots as PDF when possible, and raster task images as PNG. Keep a documented generation path for assets produced from code. Use stable descriptive filenames and labels. Place the figure and its caption in the section that argues from it.

Do not tune layout with repeated negative `\vspace` commands while the content is moving. Settle the figure, caption, and section order first. Local spacing patches are difficult to maintain and often break under another venue style.

7.5 Compile as part of writing

Compile after each meaningful change. A practical loop is:

1. run `latexmk` with errors treated as failures;
2. inspect undefined references, citations, missing files, and boxes that exceed the margin;
3. render the PDF and inspect every page at final size;
4. repeat after changes to figures, tables, title blocks, or build modes.

Before submission, build from a clean directory. Remove TODOs, author comments, and revision colors. Treat missing references and undefined citations as errors. Confirm that the source bundle contains every figure and bibliography file. A successful compiler exit is necessary, but it does not catch a clipped label or an unreadable plot. It also cannot tell you that a page is empty or a title crosses the margin.

8 Use the included LaTeX template

The source for this guide retains the preamble used by OGPO [Patil et al., 2026]. It is a working paper archive, not a generic LaTeX package. That distinction matters. The template contains reusable theorem, reference, color, and editing machinery, but it also contains OGPO-specific algorithm names and notation. Keep the former. Replace the latter before starting a new paper.

Begin with a clean copy of the project. Replace the files in `body/` and `appendix/`, clear the bibliography, and update the title block in `latex-template-main.tex`. Then remove unused definitions from `specific_macros.tex` and `algnames.tex`. A copied macro that never appears in the new paper is not harmless: it makes collisions and stale terminology harder to find.

8.1 Separate preamble files by purpose

The include order lives in `preamble/_preamble_includes.tex`. Keep that order until there is a specific reason to change it. Each file controls a different part of the build:

`color_defs.tex`

Defines link colors, theorem colors, emphasis colors, and the `Albox` environment. It loads before files that use those colors.

`header.tex`

Loads common packages and defines page geometry, citations, cross references, algorithms, and the arXiv-dependent heading commands.

`theorem_styles.tex`

Defines theorem-like environments and their appearance.

`characters.tex` and `general_macros.tex`

Define reusable mathematical characters, delimiters, operators, and formatting commands.

`specific_macros.tex`

Holds notation that belongs to one paper. This file should change substantially when the project changes.

`algnames.tex`

Defines method names, method colors, task names, and the algorithm font. Treat it as a paper-specific legend.

`color-edits.sty` and `comment_macros.tex`

Define temporary author edits, comments, deletion marks, and drafting flags.

`arxiv_title.tex`

Defines the public title layout and metadata commands for websites, code, documentation, models, and dates.

The preamble is easiest to maintain when each new command goes into the narrowest file whose purpose matches the command. A task name does not belong in `header.tex`. A package import does not belong in `specific_macros.tex`.

8.2 Use the theorem environments that are actually defined

With `customthms` enabled, choose an environment by the kind of statement:

- Use `theorem`, `lemma`, `corollary`, `proposition`, or `claim` for results.
- Use `definition`, `assumption`, or `condition` for definitions and conditions.
- Use `remark` or `example` for commentary and examples.

The template defines several less common environments as well. Use the full names `definition` and `theorem`; it does not define `defn` or `thm`.

```
\begin{definition}[Stable rollout]\label{def:stable-rollout}
A rollout is stable if ...
\end{definition}
```

```
\begin{theorem}[Main guarantee]\label{thm:main}
Under \Cref{def:stable-rollout}, ...
\end{theorem}
```

By `\Cref{thm:main}`, the error ...

Put the label immediately after the opening environment. Use `\Cref` at the beginning of a sentence and `\cref` within one. The template already names equations, assumptions, conditions, and components for `cleveref`. Equations are numbered by section.

Starred forms such as `theorem*`, `definition*`, and `remark*` suppress numbering. Use them only when no later sentence will need to cite the statement. `thminformal` provides a numbered informal theorem. An informal statement should still match the assumptions and conclusion of its formal counterpart.

The template also contains helpers for lettered variants of an existing result. For example:

```
\begin{thmmod}{thm:main}{b}{Finite-sample form}
...
\end{thmmod}
```

This prints a theorem number based on `thm:main` with the suffix `b`. Related helpers include `lemmod`, `propmod`, `defnmod`, `asmmod`, `cormod`, and `condmod`. Use them only when the new statement is genuinely a named variant of an existing result. Several other inherited helpers assume definitions from the original project and are not a stable public interface. Compile a minimal example before relying on one.

When `customthms` is false, the template does not define the full set of environments. That branch assumes the venue style provides them. Do not turn the toggle off in a plain `article` build and expect an identical theorem system.

8.3 Choose build toggles by their effect

The toggles appear near the top of `latex-template-main.tex`:

```
\newtoggle{arxiv}
\toggletrue{arxiv}
\newtoggle{customthms}
\toggletrue{customthms}
```

`arxiv` controls the public title, geometry, fonts, page style, algorithm packages, and colored paragraph heads. The false branch is a hook for a venue build, not a complete venue style by itself. In particular, the template loads `algorithm` and `algorithmic` only in the arXiv branch. A conference style must supply compatible algorithm support before the same source will compile.

`customthms` controls the colored theorem definitions. The `toreturn` toggle controls the visibility of return notes; those notes use `\RETURNN`. The inherited `icml` and `colt` toggles should remain only if the new paper still has those build modes. Remove a dead toggle rather than leaving a branch no one compiles.

The arXiv title macros also exist only when the arXiv title file is loaded. Calls such as `\paperdate` and direct access to `\metadatalist` must therefore stay inside the public build or receive venue-side definitions. The same rule applies to `\fancyhead`, because `fancyhdr` is loaded in the arXiv branch.

8.4 Choose paragraph and emphasis commands by their effect

Three commands make emphasis change with the build mode:

`\colorpar{Heading.}`

Creates a paragraph heading in both modes. The heading is colored in the arXiv build and plain in the venue branch.

`\togglepar{Heading.}`

Calls `colorpar` in the arXiv build, but becomes inline bold text in the venue branch. Use it for a compact run-in heading that can lose the paragraph command under a tight style.

`\colorbold{Text}`

Produces colored bold text in the arXiv build and ordinary bold text in the venue branch.

```
\togglepar{Experimental setup.} We compare ...
```

```
\colorpar{Failure under distribution shift.}
The policy fails when ...
```

```
\colorbold{Main finding.} The intervention ...
```

The period belongs inside the heading argument. Do not use these commands to make every paragraph look important. A paragraph heading should expose a real change in purpose: setup, result, mechanism, boundary, or implication.

`\termbold` and `\takeawaybold` use fixed semantic colors rather than a build toggle. They are appropriate only when the chosen venue permits that color. The paper must remain understandable when all emphasis is printed in grayscale.

8.5 Define method names and colors once

`algnames.tex` pairs a typographic method name with a stable color. The base helpers are:

```
\newcommand{\algofont}[1]{%
  {\scalefont{1.0}{\texttt{\textbf{#1}}}}}
\newcommand{\algofonttitle}[1]{%
  {\scalefont{1.08}{\textbf{\texttt{#1}}}}}
```

For a new method, define the color, a neutral name, a colored prose name, and a title form:

```
\definecolor{methodcol}{HTML}{A60000}
\newcommand{\Methodcolor}{methodcol}
\newcommand{\Methodnc}{\algofont{METHOD}}
\newcommand{\Method}{\color{\Methodcolor}\Methodnc}
\newcommand{\Methodtitle}{%
  {\color{\Methodcolor}\algofonttitle{METHOD}}}
```

Use `\Method{}` in prose so the following space is not swallowed. Use the neutral form when color would confuse a comparison or when the venue requires black text. Reserve the title form for a title or compact display; its font is deliberately larger.

Keep the same method-to-color assignment in plots and diagrams. Store the hex value in the plotting code as well as the LaTeX source. Color should help a reader track a method, but the method name, line style, or marker must still identify it without color.

The inherited file contains several aliases for the same method. It does not apply the neutral, colored, and title forms uniformly. Do not copy that inconsistency into a new method. Pick one naming pattern and remove aliases unused by the paper.

8.6 Keep colors semantic

`color_defs.tex` defines four semantic color groups. Link and citation colors use dark teal. Theorem headings use dark red. Term and takeaway macros use stronger red accents. Method colors live in `algnames.tex`.

Change the semantic definition instead of scattering raw color commands through the manuscript. For example, changing `\thmcolordark` should update every theorem heading. It should not require a search through theorem prose.

The inherited names are not all standard color names. Inspect the actual `\definecolor` value before reusing one. A familiar name can point to a project-specific hex code.

The environment named `AIbox` is an inherited `tcolorbox`; its name does not prescribe its content. It is used for compact summaries:

```
\begin{AIbox}{Core insight}
The critic supplies ...
\end{AIbox}
```

Use a box only when the same short statement should remain visible while the reader scans the paper. Do not place a second box inside it, and do not use a box to repeat a contribution already stated in a heading and figure.

8.7 Use color edits only during review

`comment_macros.tex` registers each author:

```
\addauthor{ms}{magenta}
\addauthor{xy}{blue}
```

Each registration creates four commands:

`\xyedit{new text}`

Shows the text in the author's color. With the `suppress` option, the text remains and loses its color.

\xycomment{question}

Shows a bracketed author comment. The `suppress` option hides it.

\xymargincomment{question}

Adds a colored marker and small comment at that point. Despite the inherited name, the current implementation does not place the full text in a conventional margin note.

\xydelete{old text}

Marks a deletion. The `showdeletions` option displays struck-out text; the default displays a marker and deletion note.

Set package options where `color-edits.sty` is loaded:

```
\usepackage[showdeletions]{preamble/color-edits}
% or, for a clean diagnostic build:
\usepackage[suppress]{preamble/color-edits}
```

A suppressed build is a diagnostic, not a final cleanup. It can hide comments, deletion markers, and return notes that still remain in the source. Before release, remove resolved edit commands and search for author prefixes, `\REF`, `\CITE`, `\TODO`, and `\RETURNN`. `\ignore` removes its argument in every build and should not hold text that might need to return.

8.8 Use the algorithm environment with its matching package

The arXiv branch uses the older `algorithmic` command set. Its commands are uppercase:

```
\begin{algorithm}[t]
\caption{Critic update from replay data}\label{alg:update}
\begin{algorithmic}[1]
\REQUIRE Initial policy and replay buffer
\FOR{each update}
  \STATE Sample a batch
  \STATE Update the critic
\ENDFOR
\end{algorithmic}
\end{algorithm}
```

`\algcomment` provides the blue monospace comments used in the OGPO pseudocode. Use it only for short operational annotations. A paragraph of explanation belongs in the caption or surrounding text.

Do not mix `algorithmic`, `algpseudocode`, and `algorithm2e` commands. They expose similar names with different semantics. If a venue supplies its own algorithm package, adapt the source once and compile both modes.

8.9 Use notation shortcuts without hiding the mathematics

`characters.tex` creates families of common symbols. Use `\bA` for a bold matrix and `\bx` for a bold vector. The same file provides `\bbR` for blackboard-bold R and `\cF` for calligraphic F .

`general_macros.tex` defines paired delimiters. These include `\abs`, `\prn`, and `\nrm`. It also defines operators such as `\argmin` and `\argmax`.

These generated names are convenient but broad. Check for an existing command before adding a new one. Do not redefine a familiar LaTeX command merely to save one character. When a symbol has paper-specific meaning, define a semantic name in `specific_macros.tex` instead of relying on a visual alias throughout the manuscript.

8.10 Publish only useful title metadata

The public title file can collect metadata below the title card:

```
\paperwebsite{https://project.example}  
\papercode{https://github.com/group/project}  
\paperdocs{https://project.example/docs}  
\papermodels{https://huggingface.co/group/project}  
\paperdate{\today}
```

`papercode`, `paperblog`, and `papermodels` include logo files from `logos/`. Keep those assets in the source bundle or replace the commands with plain links. Do not advertise an empty repository, private artifact, or placeholder URL.

The title card prints the entries in the order the commands are called. Include only metadata that helps a reader verify, reproduce, or use the work. The venue build should not depend on these public-title commands unless its style defines them.

8.11 Audit the template before submission

Build the public and venue modes from a clean directory. For each mode, check:

1. every theorem environment exists and every cross-reference resolves;
2. paragraph heads have the intended spacing after the toggle changes;
3. algorithm floats compile under the selected package;
4. method names retain the same spelling and color assignment;
5. no author edit, deletion marker, return note, or placeholder remains;
6. public metadata points to released artifacts; and
7. the source bundle contains its logos, figures, bibliography, style files, and any venue-owned class or bibliography style.

Finally, remove unused packages and source-specific macros. The preserved OGPO preamble is useful because its decisions remain visible. It becomes a burden when a new paper carries all of those decisions without a clear need for them.

9 Revise in separate passes

A single revision pass rarely works. Argument, evidence, and prose require different kinds of attention. Revise from the largest decision to the smallest. Shreyas compares the last prose pass with the final denoising step in diffusion: it can sharpen a structure that already exists, but it cannot supply a missing argument.

9.1 Pass 1: argument

Read only the material visible to a skimming reader. Start with the title and abstract, follow the section headings and figure captions, and finish with the conclusion. Write down the central claim implied by that path. Compare it with the paper contract from [Section 1](#). If they differ, fix the claim hierarchy and section order before editing prose.

Check that each section answers one question and that the sequence is necessary. Remove duplicate motivation and contribution lists. Move secondary material out of the path of the main result.

Test the argument on three readers. A student should understand the problem and its importance. An expert in the paper's field should see the technical distinction and novelty. A reviewer or expert from a neighboring field should see the broader intellectual consequence. If the framing works for only one of them, choose an intentionally narrow audience or revise the argument for broader readers.

9.2 Pass 2: evidence

Trace each major claim to the evidence that supports it. Verify every number and citation against the source. Check assumptions and uncertainty separately. A mechanism statement needs mechanism evidence; a performance statement needs a fair comparison.

Record missing controls and unresolved logic plainly. This pass may weaken a sentence or narrow a contribution. That is a correction, not a stylistic loss.

9.3 Pass 3: sections and paragraphs

Ensure each paragraph helps the reader follow one inference. Keep required setup close to the relevant result. Move implementation detail, complete proofs, and secondary sweeps to the appendix when the main argument remains self-contained.

Check transitions by reading the last sentence of one paragraph and the first sentence of the next. The first should create or answer a real question. Add a transition only when the logical connection is missing; do not add a generic recap.

9.4 Pass 4: sentences and voice

Make the minimum effective line edit. Keep the writer's vocabulary and uncertainty. Preserve the cadence when it carries the argument. Leave strong sentences alone. Untangle a long sentence when it carries several inferences, but keep it when its clauses form one clear relation.

Cut throat-clearing and inflated claims. Remove qualifiers that add no meaning; replace abstract importance with a concrete result. Prefer a direct verb to a weak verb phrase. Repeat the correct technical term instead of rotating synonyms for variety.

Watch for canned contrasts and rhetorical question-and-answer setups. Repeated paragraph shapes can sound equally mechanical. These habits flatten distinct arguments into the same rhythm. Preserve a precise contrast or short sentence when it carries real scientific or personal force.

9.5 Pass 5: the rendered paper

Read the compiled PDF alongside the source. Inspect every page at normal zoom. Check the figures and tables first, then inspect equations and links. Make sure headings do not strand a line at the bottom of a page and that floats appear near the text that interprets them.

Then read the paper aloud or at speaking pace. This catches buried subjects, repeated syntax, and sentences whose grammar works but whose logic does not. The final manuscript should sound like a researcher explaining the result to a sharp colleague.

Stop when another edit would change only the surface. The last pass should leave the paper easier to verify, not merely more uniform.

References

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- Sarvesh Patil, Mitsuhiko Nakamoto, Manan Agarwal, Shashwat Saxena, Jesse Zhang, Giri Anantharaman, Cleah Winston, Chaoyi Pan, Douglas Chen, Nai-Chieh Huang, Zeynep Temel, Oliver Kroemer, Sergey Levine, Abhishek Gupta, Hongkai Dai, Paarth Shah, and Max Simchowitz. OGPO: Sample efficient full-finetuning of generative control policies. *arXiv preprint arXiv:2605.03065*, 2026. URL <https://arxiv.org/abs/2605.03065v4>.
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- Thomas T. Zhang, Daniel Pfrommer, Chaoyi Pan, Nikolai Matni, and Max Simchowitz. Action chunking and exploratory data collection yield exponential improvements in behavior cloning for continuous control. *arXiv preprint arXiv:2507.09061*, 2025. URL <https://arxiv.org/abs/2507.09061v5>.

Section	Question	Decisive evidence
Introduction		
Setup or preliminaries		
Main result or method		
Experiments or theory		
Mechanism or ablations		
Discussion		

A.4 Figure sequence

1. **Figure 1:** What is new, what changes, and why might it change?

2. **Headline evidence:** Which figure or table establishes the central result?

3. **Mechanism or theory:** Which visual or theorem resolves the next question?

4. **Boundary:** Which result shows a failure, sensitivity, or limit?

A.5 Drafting order

☐ Claim–evidence map ☐ Figures and captions ☐ Results ☐ Setup and method ☐ Introduction
☐ Abstract and title

B Final manuscript check

B.1 Argument and evidence

- ☐ The problem, gap, central claim, and boundary are explicit.
- ☐ The motivation begins with the scientific problem and its actual bottleneck, not with the nearest acronym.
- ☐ Every abstract claim appears in the main paper.
- ☐ The abstract explains the problem, central answer, decisive evidence, and conditions under which the answer holds without touring the paper's sections.
- ☐ Every introduction contribution points to visible evidence.
- ☐ Each mechanism claim has evidence that bears on mechanism.
- ☐ Hypotheses, planned tests, expected outcomes, and validated findings are labeled separately.
- ☐ Comparators, metrics, settings, budgets, and uncertainty are named.
- ☐ Limitations narrow claims where those claims first appear.
- ☐ The conclusion introduces no new result.

B.2 Structure and prose

- ☐ Each section helps the intended reader answer one question.
- ☐ Each paragraph helps the reader follow one inference.
- ☐ Every sentence gives the reader a needed claim, condition, explanation, or transition.
- ☐ Each structural rule explains the understanding or decision it supports.
- ☐ The title and every section or subsection heading accurately describe the material that follows.
- ☐ Captions and other display labels state the evidence or action they frame.
- ☐ Main subjects and verbs are easy to find.
- ☐ Opening clauses supply a real condition, cause, or contrast.
- ☐ Quantitative statements report values and settings before interpretation.
- ☐ Evidence verbs match the strength of the evidence.
- ☐ Technical terms, method names, and notation remain consistent.
- ☐ Specialized methods are explained for the intended audience before an acronym carries the argument.
- ☐ Distinctive borrowed language and framing are cited.
- ☐ The final edit preserves the authors' voice and calibrated uncertainty.

B.3 Figures and source

- ☐ Figure 1 communicates the problem, contribution, and main evidence.
- ☐ Every later figure answers one question left open by Figure 1.
- ☐ Captions and surrounding prose make the same claim.
- ☐ Labels, legends, and annotations are legible at final size.
- ☐ The top-level entry point records the manuscript and appendix order clearly.
- ☐ Venue files are unmodified and every supported mode builds.
- ☐ TODOs, comments, revision colors, and undefined references are gone.
- ☐ The clean source bundle includes every figure and bibliography file.

B.4 Rendered PDF

- ☐ Every page has been inspected at normal zoom.
- ☐ No text, table, equation, or figure crosses a margin.
- ☐ Floats appear near the prose that interprets them.
- ☐ Links, footnotes, page numbers, and appendix references work.
- ☐ The paper reads naturally at speaking pace.